

ZISHAN SHAO

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RESEARCH INTERESTS

Machine Learning System (MLSys); Large Language Models; Diffusion Models

EDUCATION

Duke University, Durham, NC Incoming Fall 2026
Ph.D. in Electrical & Computer Engineering

Duke University, Durham, NC Aug 2024 – May 2026
M.S. in Statistical Science

Wake Forest University, Winston-Salem, NC Aug 2020 – May 2024
B.S. in Computer Science with Honors & B.S. in Statistical Science, *summa cum laude*

EXPERIENCE

Research Assistant, Center for Computational Evolutionary Intelligence (CEI), Duke Fall 2024 – Present
Advisors: Prof. Yiran Chen, Prof. Hai "Helen" Li

Research Assistant, REASON Lab, Wake Forest University Spring 2022 – Fall 2025
Advisor: Prof. Aditya Devarakonda

SELECTED PUBLICATIONS

* denotes equal contribution

- **Zishan Shao**, Lixun Zhang, Kangning Cui, Yixiao Wang, Ting Jiang, Hancheng Ye, Qinsi Wang, Zhixu Du, Yuzhe Fu, Fan Yang, Danyang Zhuo, Yiran Chen, Hai Helen Li. "DecodeShare: Tracing the Shared Pathways of LLM Decode-Time Decisions." **ICML 2026** (**Spotlight, Top 2.2%**). [\[PDF\]](#)
- **Zishan Shao**, Yixiao Wang, Qinsi Wang, Ting Jiang, Zhixu Du, Hancheng Ye, Danyang Zhuo, Yiran Chen, Hai Li. "FlashSVD: Memory Efficient Approach for SVD-Based Low Rank Model Inference." **AAAI 2026**. [\[PDF\]](#)
- **Zishan Shao**, Aditya Devarakonda. "Scalable Dual Coordinate Descent for Kernel Methods." **HPCA Asia 2025** (**Outstanding Paper Award**). [\[PDF\]](#)
- Yixiao Wang*, **Zishan Shao***, Ting Jiang, Aditya Devarakonda. "Enhanced Cyclic Coordinate Descent Methods for Elastic Net Penalized Linear Models." **NeurIPS 2025**. [\[PDF\]](#)
- Yixiao Wang*, Ting Jiang*, **Zishan Shao***, Hancheng Ye, Jingwei Sun, Mingyuan Ma, Qinsi Wang, Jianyi Zhang, Yiran Chen, Hai Li. "ZEUS: Zero-shot Efficient Unified Sparsity for Generative Models." **ACM MM**, 2026. [\[PDF\]](#)
- Ting Jiang*, Hancheng Ye*, Yixiao Wang*, **Zishan Shao**, Jingwei Sun, Jingyang Zhang, Jianyi Zhang, Zekai Chen, Yiran Chen, Hai Li. "SADA: Stability-guided Adaptive Diffusion Acceleration." **ICML 2025**. [\[PDF\]](#)
- Qinsi Wang, Jing Shi, Kun Wan, Handong Zhao, Hancheng Ye, **Zishan Shao**, Jinghan Ke, Yudong Liu, Daniel Miranda, Purvak Lapsiya, Yiran Chen, Wentian Zhao. "Seeing is Solving: Unlocking Efficient Multimodal RL via View Alignment." **ICML 2026**.

SELECTED MANUSCRIPTS

- Wenhao Wu*, **Zishan Shao***, Kangning Cui, Jinhee Kim, Yixiao Wang, Hancheng Ye, Danyang Zhuo, Yiran Chen. "FlashSVD v1.5: Making Low-Rank Transformers Inference Actually Fast." *arXiv preprint arXiv:2605.08314*, 2026. [\[PDF\]](#)

SELECTED RESEARCH PROJECTS

DecodeShare: Tracing the Shared Subspace of LLM Decode-Time Decisions

[Project Page](#)

Duke Center for Computational Evolutionary Intelligence (CEI)

- Identified and causally validated a compact, task-general subspace carrying decision-relevant information in KV-cached LLM decoding through decode-only ablations and targeted patchback experiments.

FlashSVD: Unified Runtime for Low-Rank Transformer Inference

[Project Page](#)

Duke Center for Computational Evolutionary Intelligence (CEI)

- *FlashSVD V1 — Memory-Efficient Kernels Suite*: Developed rank-aware CUDA/Triton kernels that execute low-rank attention and FFN layers without reconstructing full intermediate activations, reducing peak activation memory by 70.2% and transient memory by 75%.
- *FlashSVD V1.5 — Unified Inference Runtime*: Built a common execution framework for heterogeneous SVD-compressed checkpoints, including SVD-LLM and Basis Sharing, enabling optimized inference without accuracy degradation.

ZEUS: Zero-shot Efficient Unified Sparsity for Generative Models

[Project Page](#)

Duke Center for Computational Evolutionary Intelligence (CEI)

- Developed a training-free acceleration framework for generative models using second-order extrapolation and parity-aware feature reuse, achieving $1.9\times$ – $3.6\times$ speedups with improved perceptual fidelity over prior training-free methods at matched acceleration.

SERVICE & MENTORSHIP

- **Conference Reviewer**: AAAI (2026, 2027), ICLR (2026), NeurIPS (2026; Main Track, D&B Track)
- **Workshop Reviewer**: AdaptFM @ ICML (2026)
- **Research Mentorship**: Mentored 7 Duke students (1 Ph.D., 2 M.S., 4 B.S.), contributing to 1 ICML Spotlight and 1 workshop submission.

HONORS, AWARDS & FELLOWSHIPS

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| • Cloudex GPU Catalyst Fellowship (Lab award, PI: Prof. Yiran Chen; proposal co-author)
<i>90/10 matching grant; \$10,000 compute credits (5000 H100 or 4000 H200 GPU-hours).</i> | 2026 |
| • Outstanding Paper Award , HPCAsia | 2025 |
| • Phi Beta Kappa (Academic Honor Society) & Upsilon Pi Epsilon (CS Honor Society) | 2023 |
| • Wake Forest Research Fellowship & George Washington Greene Scholarship | 2023 |
| • UPE Scholarship (Upsilon Pi Epsilon National Honor Society) | 2023 |
| • Meritorious Winner , COMAP International Mathematical Contest in Modeling (ICM) | 2022 |

SKILLS

Languages & Frameworks: Python, C/C++, CUDA, Triton, PyTorch

Systems & Optimization: Kernel fusion, memory tiling, mixed precision, GPU profiling (nsys, Nsight Compute), HPC, Linux